Chapter Twenty-Two

Muddled Policies in Data and E-Commerce

Fundamentality of Data Creation, Storage and Use for the Digital Economy

All numbers, words, pictures, sounds, information, expressions and everything else constructed in bytes are data. Data has been described as the new oil. This is an incorrect description. Data is more like matter. All the agriculture and industrial goods produced are made of matter. All digital goods and services produced are made of data. Some matter works better as fuel or energy. Oil is more of a fuel and not building block matter. For the digital economy, data does not work like fuel, semiconductors perhaps do. For the digital economy, data is everything: it is the input, the process of production and the output. The economy cannot be digitalized without extensive creation and use of data. Data is the matter for the digital economy and semiconductors are fuel.

Humans are factors of production and consumers of most goods and services produced in the economy. Digitalization offers the opportunity of a fundamental reset from the production systems of the industrial economy. The industrial economy was most competitive and productive producing goods at scale. This led to mass production of almost every product. Though humans have innumerable subtle variations in physical parameters, mass production could only afford a few. For instance, shirts could be produced only in three or four sizes. With more variations, scale and competitiveness were lost. In the process, products could not be customized precisely to each individual. Customization was a lost cause in the Industrial Revolution.